

Ex.No:9	DATA VISUALIZATION USING POWER BI
Date:	

AIM:

To import, analyze, visualize, and create interactive dashboards using **Microsoft Power BI** for effective data analysis and to understand the economic impact, vulnerability, and financial indicators of different countries.

ALGORITHM:

Country Economic Impact Visualization Using Power BI

Step 1: Data Loading

Open Microsoft Power BI Desktop.

Select Home → Get Data → Text/CSV.

Import the country_impact.csv dataset.

Verify the imported fields:

Country

Region

Oil Import %

GDP Impact %

Inflation Risk

Stock Market Change

Currency Pressure

Policy Response

Vulnerability

Population (M)

Click Load to load the dataset into Power BI.

Step 2: Country-wise Oil Import Visualization

Create a new Clustered Column Chart.

Place Country in the X-axis.

Place Oil_Import_Pct in the Y-axis.

Enable Data Labels.

Give the visual the title "Oil Import Dependency by Country".

Use the chart to compare the percentage of oil imports among different countries.

Step 3: GDP Impact Visualization

Create a new Bar Chart.

Place Country in the Y-axis.

Place GDP_Impact_Pct in the X-axis.

Enable data labels.

Give the chart the title "GDP Impact by Country".

Analyze the differences in GDP impact among countries.

Step 4: Stock Market Change Visualization

Create a Column Chart.

Place Country on the X-axis.

Place Stock_Market_Change on the Y-axis.

Enable data labels.

Use the chart to observe differences in stock market changes across countries.

Apply suitable formatting to distinguish positive and negative changes.

Step 5: Vulnerability Analysis

Create a Donut Chart.

Place Vulnerability in the Legend.

Place Country in Values and set the aggregation to Count.

Display the percentage and category labels.

Give the chart the title "Country Vulnerability Distribution".

Analyze the distribution of countries across different vulnerability categories.

Step 6: Inflation Risk Visualization

Create a Pie Chart.

Place Inflation_Risk in the Legend.

Place Country in Values and select Count.

Enable data labels.

Give the visual the title "Inflation Risk Distribution".

Observe the number of countries under each inflation-risk category.

Step 7: Region-wise Analysis

Create a Map/Column Chart or Stacked Column Chart.

Place Region on the X-axis.

Place GDP_Impact_Pct on the Y-axis.

Use Country as the legend if required.

Give the visual the title "Regional Economic Impact Analysis".

Compare the economic impact across different regions.

Step 8: Create a Calculated Measure

Select Modeling → New Measure.

Create a measure named Average GDP Impact.

Use the following DAX formula:

Average GDP Impact = AVERAGE(country_impact[GDP_Impact_Pct])

Display the measure using a Card Visual.

Give the card the title "Average GDP Impact %".

Step 9: Population Analysis

Create a Column Chart.

Place Country on the X-axis.

Place Population_M on the Y-axis.

Enable data labels.

Give the chart the title "Population by Country".

Use the chart to compare the population represented in the dataset.

Step 10: Create Interactive Filters

Add a Slicer for Region.

Add another Slicer for Country.

Add a Slicer for Vulnerability.

Select different values and observe how the dashboard visuals change dynamically.

Use these slicers to perform region-wise and country-wise analysis.

Step 11: Dashboard Creation

Create a Power BI report page containing the following visuals:

Oil Import Dependency by Country

GDP Impact by Country

Stock Market Change

Vulnerability Distribution

Inflation Risk Distribution

Regional Economic Impact

Population by Country

Average GDP Impact Card

Arrange the visuals clearly on the report canvas.

Add appropriate titles and labels.

Apply a suitable Power BI theme.

Add Country, Region, and Vulnerability slicers.

Ensure that selecting a slicer dynamically updates the other visuals.

Step 12: Analysis

Analyze the oil import dependency of different countries.

Compare GDP impact percentages among countries.

Examine stock market changes.

Identify the distribution of vulnerability categories.

Analyze inflation-risk levels across countries.

Compare economic impacts among different regions.

Observe population differences between countries.

Use slicers to dynamically analyze individual countries and regions.

Use the dashboard to understand the relationship between oil dependency, economic impact, financial indicators, and country vulnerability.

OUTPUT:

Card Creation:



Region		
Africa	Europe/Asia	Oceania
East Asia	Middle East	South Asia
Europe	North America	

Avg GDP Impact %

-3.51

Avg Stock Change %

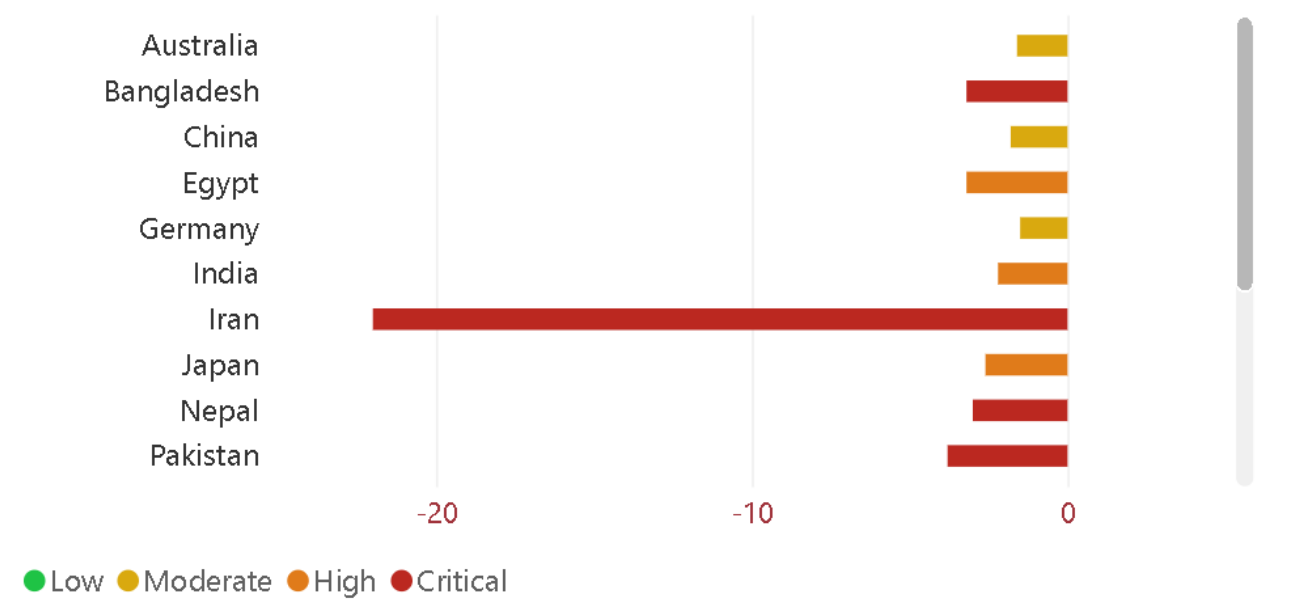
-10.00

Critical Countries

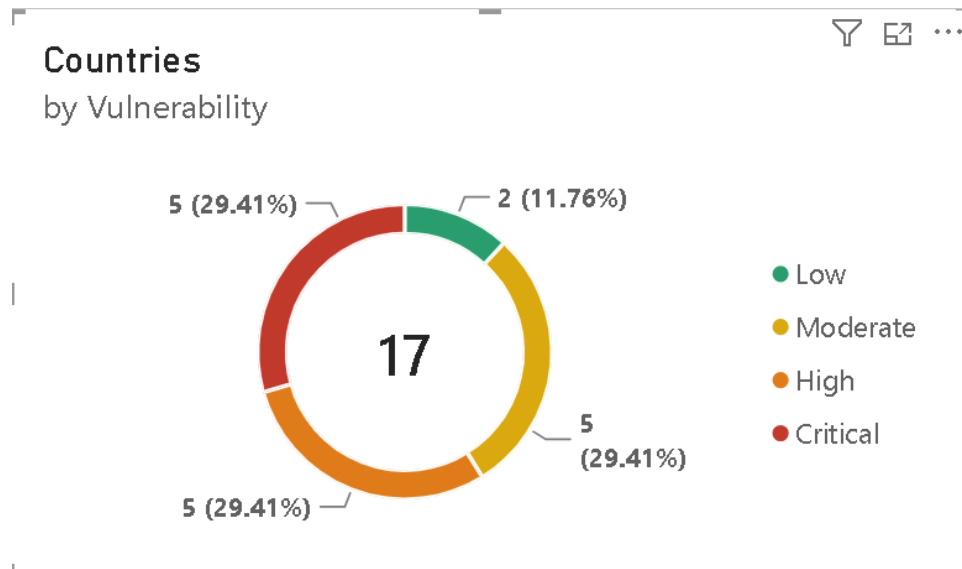
5

Bar Chart Creation:

Sum of GDP_Impact_Pct
by Country and Vulnerability



Donut Chart:



RESULT:

Successfully imported, analyzed, and visualized the **Country Impact dataset** using **Microsoft Power BI**. Created interactive charts, calculated measures, slicers, and a comprehensive dashboard to analyze **oil import dependency, GDP impact, stock market changes, inflation risk, currency pressure, population, regional differences, and country vulnerability**, enabling effective data exploration and data-driven economic analysis.

